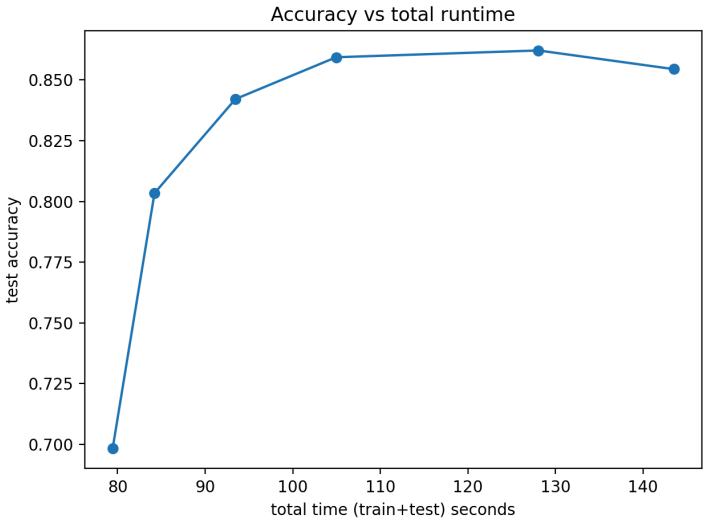
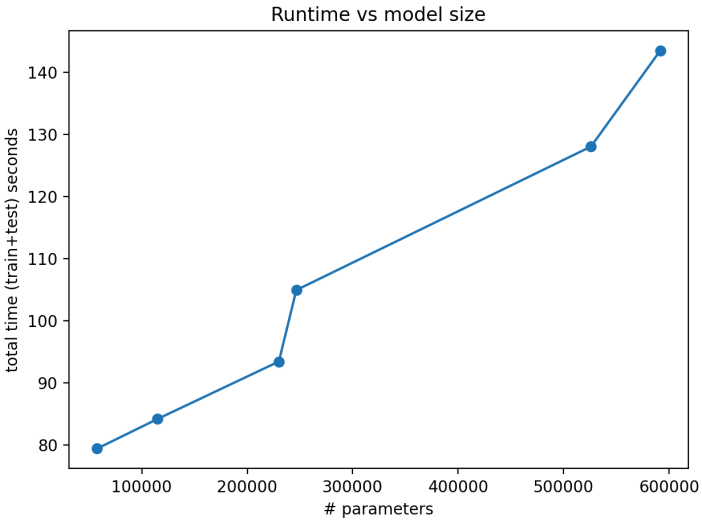
Mini Project 2B Report

By Andrew Klusmeyer & Kyle Zink

Without GPU Enabled

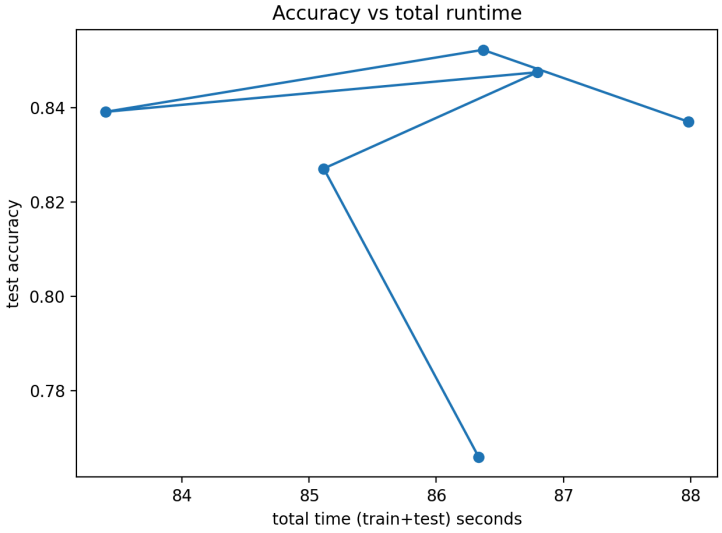
Model	Hidden Layers	Params	Train Acc	Test Acc	Train Time (s)	Test Time (s)	Total Time (s)
baseline_h32	[32]	57,508	0.962	0.698	57.3	22.1	79.4
m1_h64	[64]	115,012	0.996	0.803	61.4	22.8	84.2
m2_h128	[128]	230,020	0.998	0.842	69.9	23.5	93.5
m3_h128x2	[128, 128]	246,532	0.999	0.859	80.7	24.3	105.0
m4_h256x2	[256, 256]	525,828	1.000	0.862	101.8	26.2	128.0
m5_h256x3	[256, 256, 256]	591,620	0.999	0.854	116.7	26.8	143.5

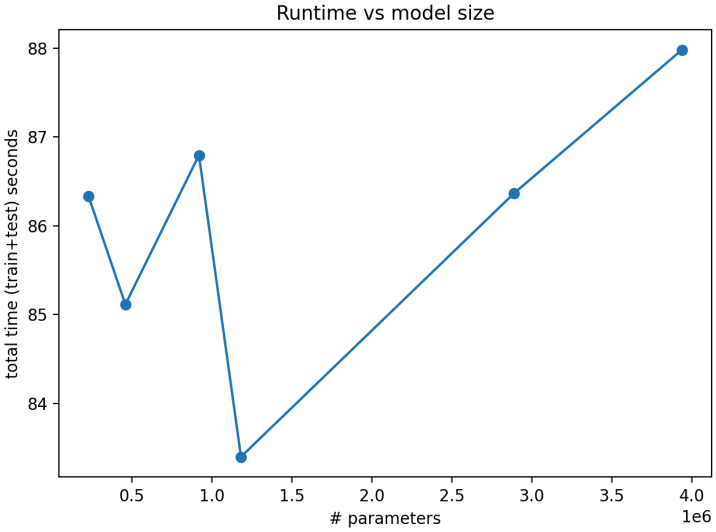




With GPU Enabled

Model	Hidden Layers	Params	Train Acc	Test Acc	Train Time (s)	Test Time (s)	Total Time (s)	Device
baseline_h128	[128]	230,020	0.977	0.766	47.0	39.3	86.3	cuda
m1_h256	[256]	460,036	0.996	0.827	45.6	39.5	85.1	cuda
m2_h512	[512]	920,068	0.997	0.847	47.1	39.7	86.8	cuda
m3_h512x2	[512, 512]	1,182,724	0.993	0.839	45.8	37.6	83.4	cuda
m4_h1024x2	[1024, 1024]	2,889,732	0.998	0.852	46.9	39.5	86.4	cuda
m5_h1024x3	[1024, 1024, 1024]	3,939,332	0.994	0.837	48.3	39.7	88.0	cuda





Summary of Results

Using the full GAMEEMO dataset (128-sample windows, subject-wise split, 100 epochs), test accuracy increased with model size and then plateaued. On the FABRIC node (8 CPU cores, 16GB RAM, RTX 6000 GPU, ~100GB storage), the CPU-only sweep improved from ~0.70 to a peak near ~0.86 but runtime rose substantially as models grew. With GPU enabled, we trained much larger models (up to ~3.9M parameters) while runtimes stayed roughly flat (~46–48s train, ~38–40s test), and accuracy peaked around ~0.85 before slightly declining at the largest size, indicating diminishing returns.